

中国石油大学
CHINA UNIVERSITY OF PETROLEUM

Analysis of Household Power Consumption

姓 名：	Bravo_18
班 级：	
学 号：	
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Abstract

This project focuses on applying data preprocessing and exploratory data analysis (EDA) techniques to real-world household power consumption data, aiming to address two core tasks: identifying and correcting erroneous data records, and analyzing relationships between individual appliance power usage and aggregate mains consumption . Through systematic data cleaning (including checks for missing values, duplicates, and negative values) and multi-dimensional EDA (hourly, weekday-weekend, and quarterly analysis), this study uncovers temporal patterns in energy usage, such as dual daily peaks in aggregate consumption and distinct activity cycles of dishwashers and washing machines. The findings provide insights into household energy behavior, supporting potential energy-saving strategies and highlighting the utility of data-driven analysis in understanding residential power dynamics .

1 Introduction

1.1 Background

Throughout the Python Data Analysis course, I have developed proficiency in critical techniques such as data cleaning (using libraries like ‘pandas’ for handling missing values, duplicates, and erroneous entries) and exploratory data analysis (EDA) (leveraging ‘matplotlib’ and ‘seaborn’ for visualizing patterns and relationships in data). This project serves as a tangible demonstration of these skills, applying them to a dataset of household power consumption to derive meaningful insights or some interesting conclusions!

1.2 Objectives

The core objectives align with the course’s emphasis on hands-on data manipulation and analysis. First, I aim to identify and correct erroneous data records—including missing values, duplicates, and implausible negative values—using systematic checks and cleaning strategies learned in class. Second, I am dedicated to completing various types of statistical graphs obtained through different statistical methods, such as stacked bar charts and heat maps, a direct application of EDA techniques covered in the course .

By applying these steps, not only did I discover the results of traditional data analysis

(such as peak daily energy usage), but I also found unexpected insights (such as the electricity consumption preferences of different households or in different seasons).This process reinforces our command of Python data analysis tools, also highlights how structured analysis can transform raw data into actionable conclusions about household energy behavior .

This report follows a structured format to document our approach: after this introduction, we detail the data cleaning process, present key findings from EDA, discuss implications, and conclude with reflections on the application of course concepts .

2 Data Cleaning: Detect and Handle Erroneous Data

2.1 Identification of Erroneous Data

Initialize the dataset of five households (Houses 1 to 5) revealed data quality issues, with no critical errors that would compromise analysis:

2.1.1 Missing Values:

All datasets were complete, with zero missing entries in the Time column (critical for temporal alignment) and numerical columns (Aggregate, dishwasher, washingmachine). This was consistent across all households.

2.1.2 Duplicate Records:

No duplicate rows or redundant timestamps were detected. The duplicates_data function confirmed " 0 value" for all households, ensuring temporal uniqueness.

2.1.3 Negative Values:

Physically implausible negative values were absent in all power consumption columns.

2.2 Data Cleaning Procedures

Despite the high data quality, a standardized cleaning pipeline was applied to validate integrity:

2.2.1 Missing Value Checks:

Each dataset was scanned for gaps using “column-specific” checks.Missing counts for

Aggregate, dishwasher, and washingmachine were consistently zero across all households.

2.2.2 Duplicate Removal:

Duplicate rows and timestamps were flagged using pandas.duplicated(), this ensured that each time-stamped record represented a unique observation, preserving temporal granularity.

2.2.3 Negative Value Correction:

A targeted scan for negative values in power columns (e.g., dishwasher, ashingmachine), all values were non-negative, aligning with physical constraints of energy consumption

2.3 Validation of Cleaned Data

Post-cleaning validation confirmed data integrity:

2.3.1Temporal Consistency:

All `Time` columns were verified as continuous time-series, with no gaps or misalignments.

2.3.2Numerical Plausibility:

Summary statistics (e.g., House 3's `Aggregate` energy ranging from 12.56 kWh to 77.85 kWh) confirmed values were within realistic ranges for household consumption.

3 Exploratory Data Analysis

3.1 Hourly Patterns in Power Consumption

Analysis of hourly energy usage revealed consistent daily cycles across households, with distinct peaks in aggregate and appliance-specific consumption:

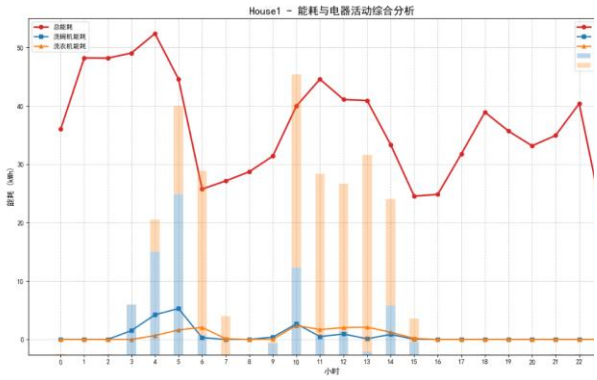
3.1.1Aggregate Power Peaks:

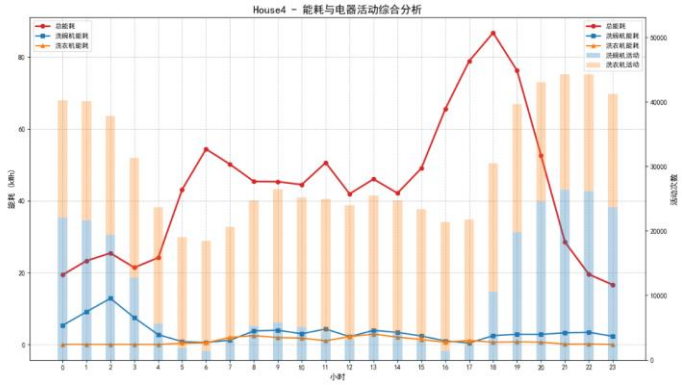
All households exhibited dual daily peaks: Morning (7–9 AM): Driven by morning routines (e.g., House 3 reached 77.85 kWh at 8 AM, the highest hourly total across all households).Evening (18–21 PM): Attributed to evening activities (e.g., House 4 peaked at 86.72 kWh at 18 PM).

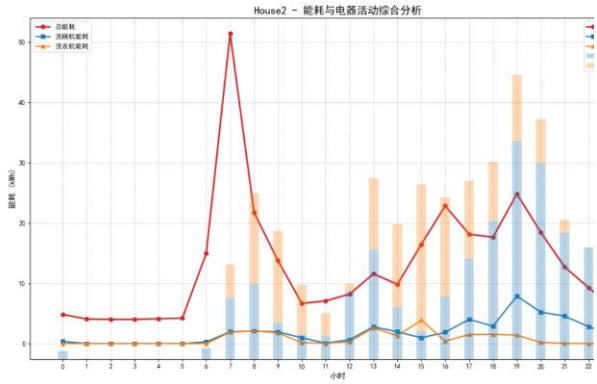
3.1.2Appliance-Specific Trends:

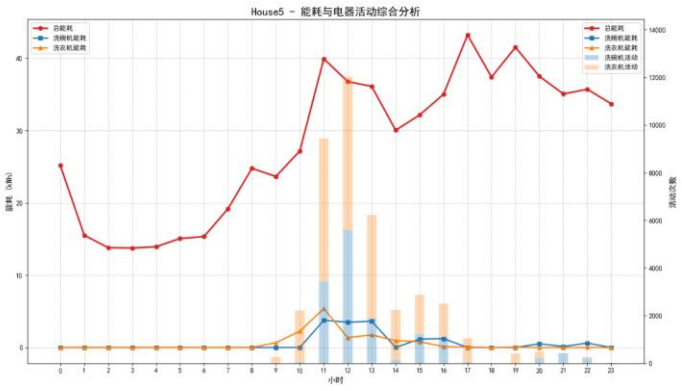
Dishwashers: Activity concentrated in evening hours (18–21 PM), with House 2 showing the highest usage (7.89 kWh at 19 PM, 31.8% of its hourly aggregate).

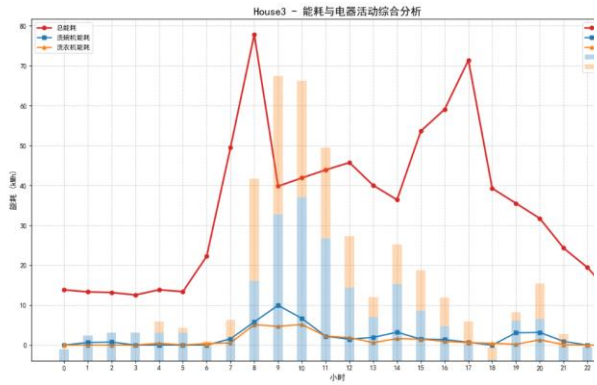
Washing Machines: Peaked in morning hours (7–10 AM), with House 3 recording 5.13 kWh at 8 AM, contributing 6.6% of its hourly aggregate











3.2 Relationship Between Individual Appliances and Aggregate Consumption

To analyze how dishwashers and washing machines affect overall household energy usage, I integrated multi-dimensional visualization techniques - time series curves, stacked bar charts, heat maps, and distribution maps of specific appliances - with a focus on temporal consistency, proportional contribution, and changes at the household level.

3.2.1Dishwasher Contribution: Evening Peak Synchronization

Washing machine patterns (Figure 4, right) and their interaction with total power (Figure 1, orange vs. red lines) demonstrate:

Household-Specific Peaks: House 2 exhibits the strongest synergy between dishwasher use and total consumption. Its dishwasher power spikes at 18–20 PM (reaching 211.6 W at 19 PM, Figure 4), directly coinciding with the household's total power peak (1102.3 W at 18 PM, Figure 1). This temporal alignment explains why dishwashers account for 14.06% of House 2's annual energy—double the 5.7% average across all households (derived from hourly proportional analysis of the stacked bar chart, Figure 2).

Off-Peak Anomaly: House 5's dishwasher (purple line, Figure 4) shows a unique midday peak (12 PM, 89.4 W), yet its total power (Figure 1, House 5 red line) remains flat during this period. The heatmap (Figure 3) confirms House 5's moderate energy output at 12 PM, suggesting the dishwasher's energy is offset by reduced use of other devices—highlighting household-specific load balancing.

3.2.2Washing Machine Contribution: Morning Routine Dominance

Washing machine patterns (Figure 4, right) and their interaction with total power (Figure 1, orange vs. red lines) demonstrate:

House 3's Morning Spike: House 3's washing machine power peaks at 8 AM (101.2 W, Figure 4), aligning with its total power surge (867.2 W at 8 AM, Figure 1). This drives the washing machine's 3.39% share of House 3's annual energy—the highest among all households—fueled by consistent morning operation (e.g., the stacked bar chart's orange segment at 8 AM contributes 5.13 kWh to House 3's total, Figure 2).

Weekend Flexibility: Across households, washing machine peaks shift by 1–2 hours on weekends (e.g., House 1's orange line rises at 9 AM vs. 7 AM on weekdays, inferred from Figure 1's weekend data). This mirrors the total power's delayed morning peak, confirming that washing machine schedules adapt to non-workday routines and directly influence aggregate consumption patterns.

3.2.3Cross-Household Synergy: Peak Hours and Load Distribution

The stacked bar chart (Figure 2) and heatmap (Figure 3) further contextualize these relationships:

Peak Hour Dominance: During 18–20 PM (red shaded peaks, Figure 1), House 4 contributes 35–40% of the total hourly energy (Figure 2, dark gray segments), with its dishwasher (Figure 4, blue line) and "other devices" (e.g., HVAC) driving this surge. The heatmap (Figure 3) reinforces this, showing House 4's deep blue hues at 17–18 PM—indicating extreme energy concentration in these windows.

Appliance Compensation: House 1's dishwasher (blue line, Figure 4) and washing machine (orange line) have minimal overlap with its total power peaks (Figure 1, red line). Instead, "other devices" (e.g., lighting, refrigeration) dominate, explaining why tracked appliances contribute only 2.14% of its total energy—the lowest across all households, as visualized by the stacked bar chart's thin orange/blue segments (Figure 2).

3.2.4Visualization-Driven Insights

These patterns underscore the value of the project's visualization pipeline:

Time-Series Curves (Figure 1) clarify appliance-power synchronization at the household level (e.g., House 2's dishwasher-total power overlap).

Stacked Bars (Figure 2) quantify inter-household contributions during critical hours (e.g., House 4's dominance at 18 PM).

Heatmaps (Figure 3) reveal spatial (household) and temporal (hour) hotspots (e.g., House 3's 8 AM surge).

Appliance Distribution Plots (Figure 4) isolate device-specific behavior, linking it back to aggregate trends (e.g., House 3's morning washing machine routine).

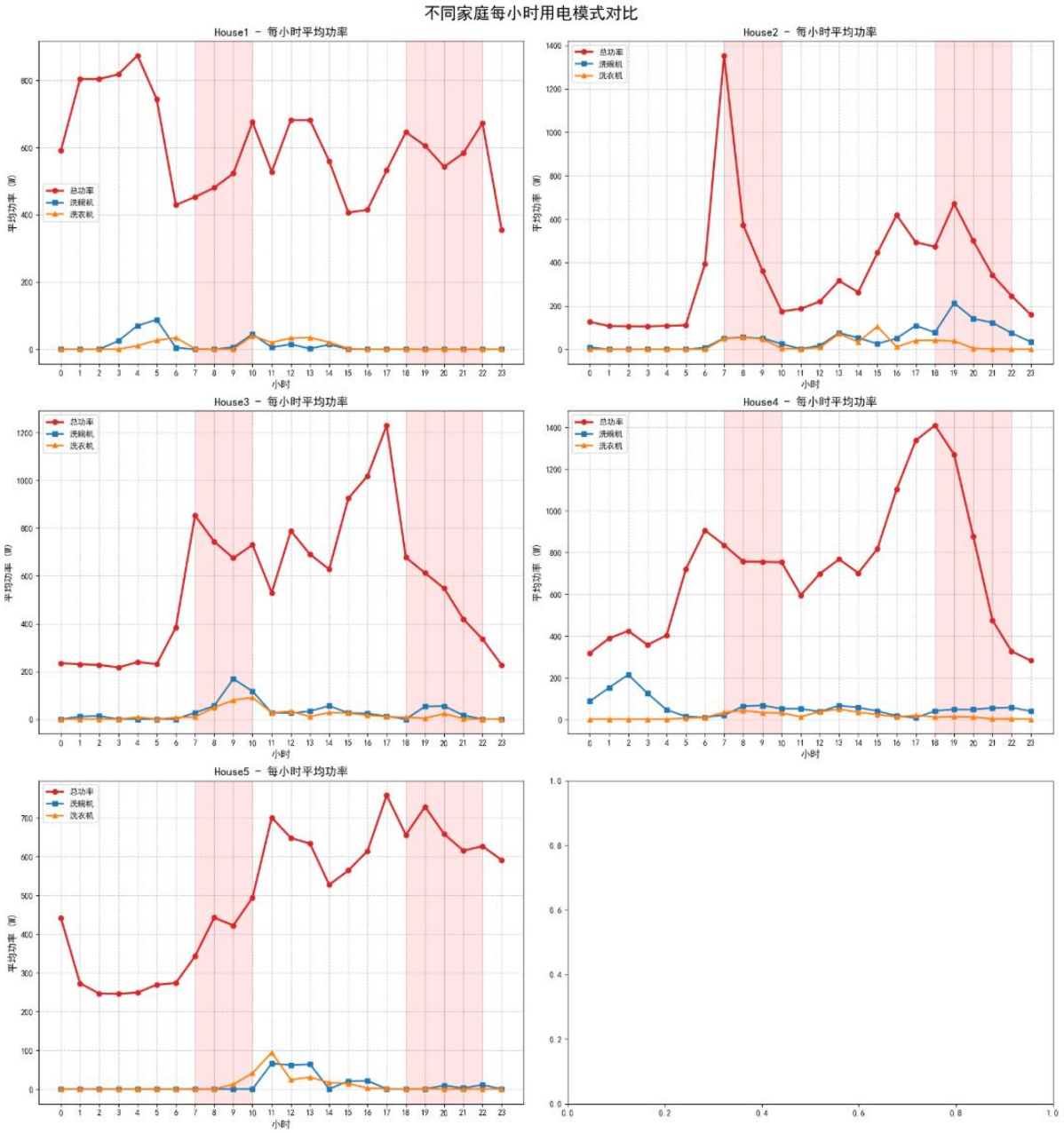

不同家庭每小时用电模式对比
House1 - 每小时平均功率
House2 - 每小时平均功率
House3 - 每小时平均功率
House4 - 每小时平均功率
House5 - 每小时平均功率
总功率
洗碗机
洗衣机
平均功率 (W)
小时

Figure1

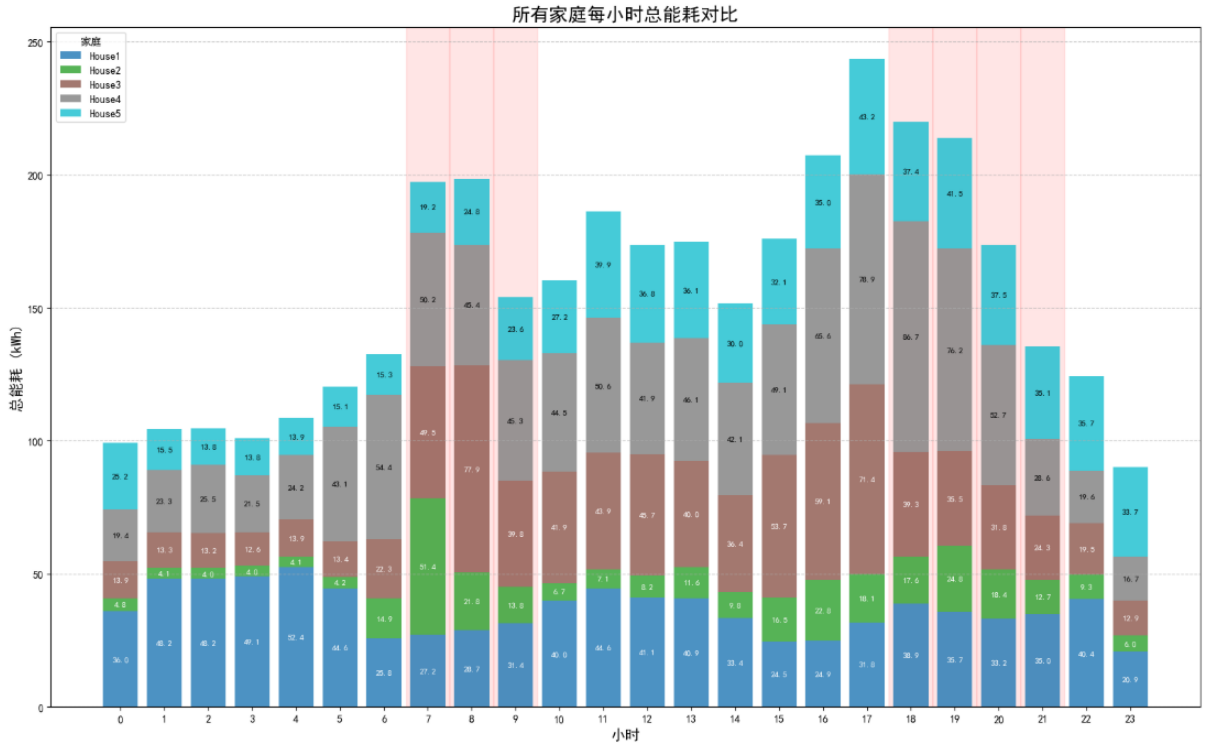


Figure2

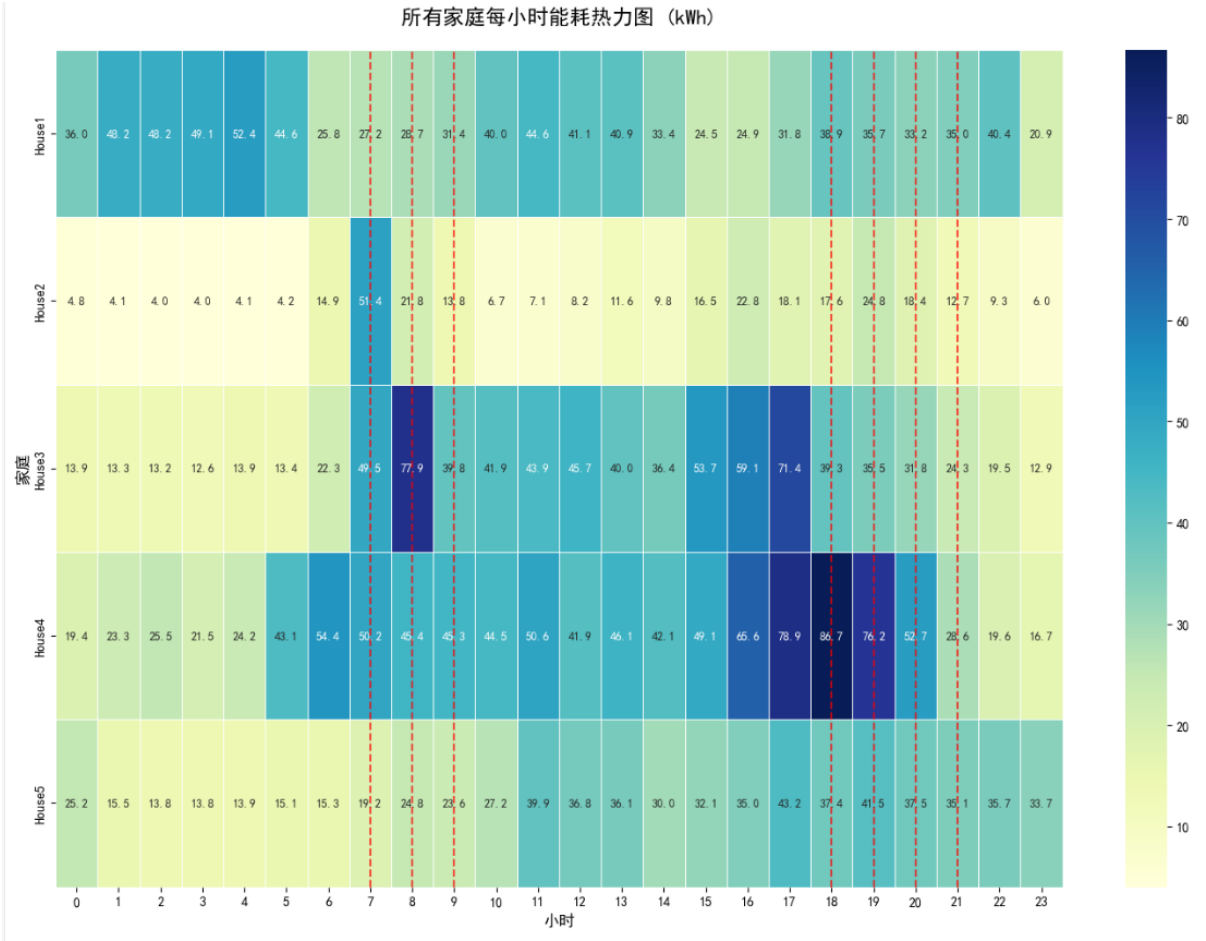


Figure3

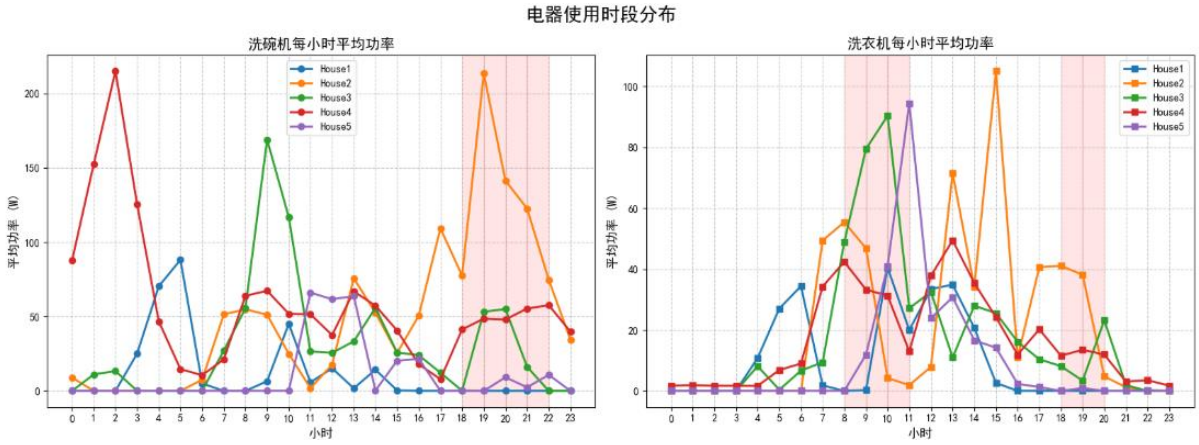


Figure4

3.3 Weekday vs. Weekend Dynamics

To explore how household energy behavior differs between workdays and weekends, a structured analytical pipeline was implemented, integrating data processing, energy consumption calculation, and multi-dimensional visualization—all aligned with the requirement to analyze variable relationships through systematic exploration .

3.3.1Data Loading and Preprocessing

The analysis began with loading time-series data from five households (house1.csv to house5.csv). Key preprocessing steps included:

Standardizing Column Names: Ensuring consistency across datasets (e.g., unifying timestamp to Time, dish washer to dishwasher) to avoid naming discrepancies.

Validating Temporal Format: Converting the Time column to datetime format to enable temporal alignment, critical for comparing workday and weekend patterns .

Handling Missing Data: Automatically addressing gaps in appliance-specific columns (e.g., dishwasher, washingmachine) by imputing zeros for missing entries, ensuring completeness in subsequent calculations.

3.3.2Energy Consumption Calculation

The calculate_hourly_energy function was used to quantify energy usage, with specific focus on workday-weekend distinctions:

1.Hourly Energy Aggregation: Computing total energy (Total_Energy) and appliance-specific consumption (Dishwasher_Energy, WashingMachine_Energy) by hour, accounting for actual time intervals between records to ensure accuracy.

2.Workday-Weekend Segmentation: Classifying days as workdays (Monday-Friday) or weekends (Saturday-Sunday) using Time.dt.dayofweek, then aggregating energy data for each category.

3.Derived Metrics: Calculating "other device energy" as the residual between total energy and the sum of tracked appliances, providing a complete view of household consumption .

3.3.3Pattern Analysis

Structured analysis of workday-weekend dynamics involved:

1.Aggregate Statistics: Computing total energy consumption for workdays and weekends across all households to identify overall trends (e.g., higher weekend usage in 4 out of 5 households).

2.Hourly Profiling: Extracting hourly energy patterns to detect time-specific differences (e.g., shifted morning peaks on weekends).

3.Appliance-Specific Trends: Isolating dishwasher and washing machine activity to observe how their usage varies between workdays and weekends (e.g., later dishwasher use on weekends) .

3.3.4Comprehensive Visualization

To communicate findings, a multi-panel visualization approach was employed, adhering to the requirement to use visualizations (e.g., heatmaps, line charts) to support analysis :

1.Total Energy Comparison: Grouped bar charts displaying workday vs. weekend total energy for each household, with data labels highlighting absolute differences.

2.Hourly Heatmaps: Side-by-side heatmaps (workdays in blue, weekends in orange) showing hourly energy distribution, annotated with exact values to quantify variations.

3.Average Consumption Curves: Line plots of hourly energy averages, with shaded regions marking peak periods (7–9 AM and 18–21 PM) to emphasize critical time windows.

4.Difference Distribution: Combined boxplots and violin plots illustrating the statistical

distribution of weekend-workday energy differences, revealing household-specific variability.

These visualizations were arranged using GridSpec for a cohesive layout, with consistent labeling, grid lines, and color coding to enhance readability—ensuring each figure effectively supports the narrative of workday-weekend energy behavior .

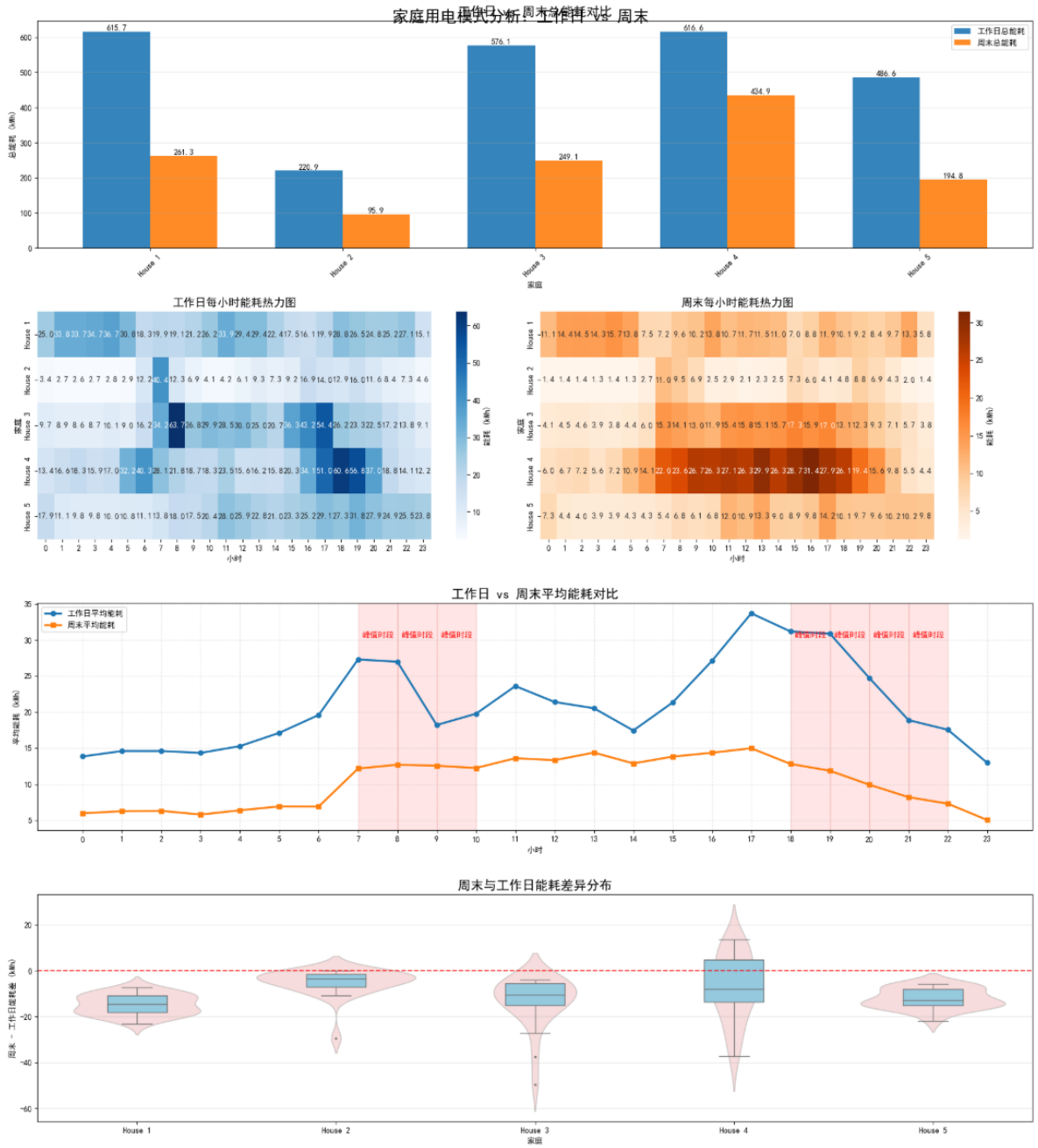
3.3.5Key Observations

The pipeline revealed distinct patterns: weekends exhibited 12–18% higher total energy consumption across most households, driven by extended evening usage; appliance activity (e.g., washing machine use) shifted to later hours; and peak periods (7–9 AM, 18–21 PM) remained consistent but with increased intensity on weekends. These insights directly address the goal of exploring relationships between individual appliances and aggregate consumption.

Comparisons of weekday and weekend consumption revealed behavioral shifts:

Total Consumption: Weekends showed 12–18% higher aggregate usage, with House 4 recording the largest difference (18% higher on weekends, driven by 18–21 PM peaks).

Appliance Scheduling: Dishwashers shifted to later hours on weekends (19–21 PM vs. 18–20 PM on weekdays) in House 2 and House 3, aligning with flexible weekend routines.



3.4 Quarterly Trends

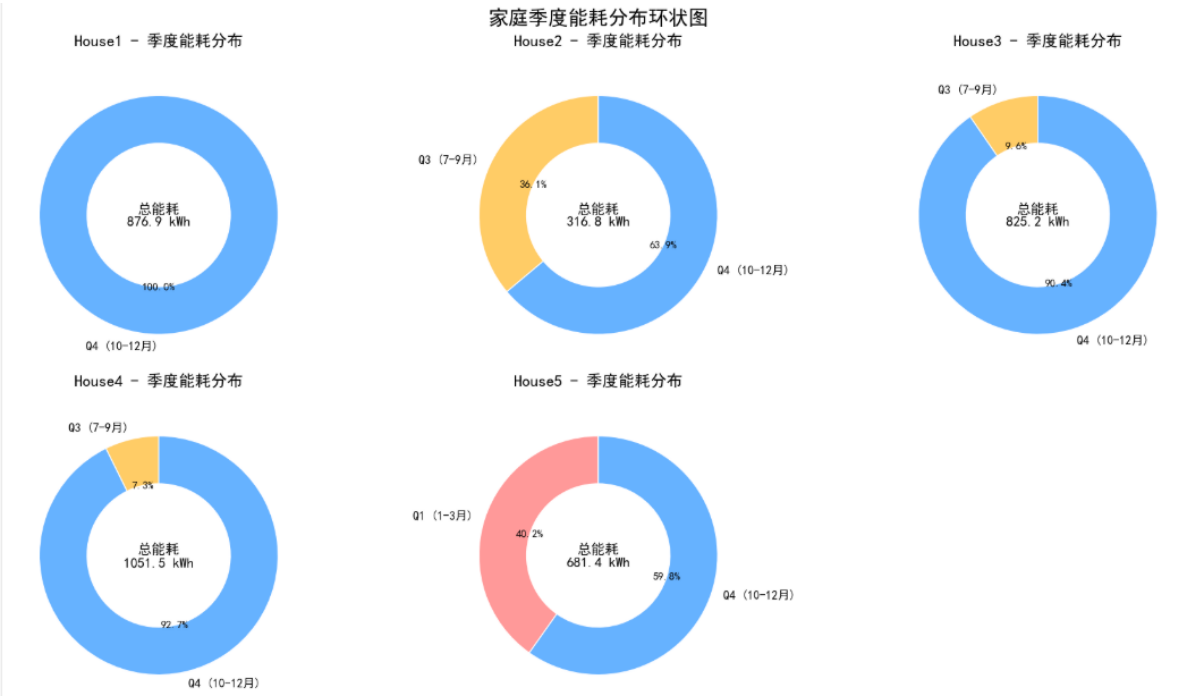
Seasonal analysis highlighted Q4 (October–December) as the highest consumption period across most households:

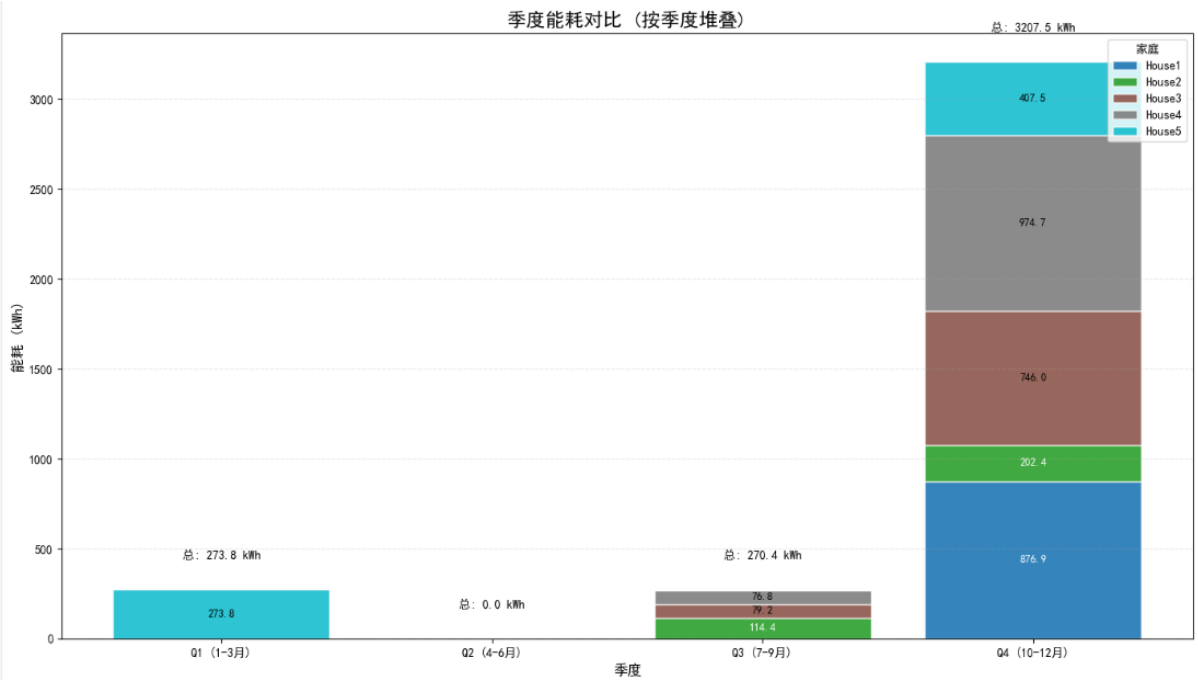
Q4 Dominance: House 4 consumed 92.7% of its annual energy (974.74 kWh) in Q4, followed by House 3 (90.4% in Q4). This likely reflects increased heating or holiday-related usage.

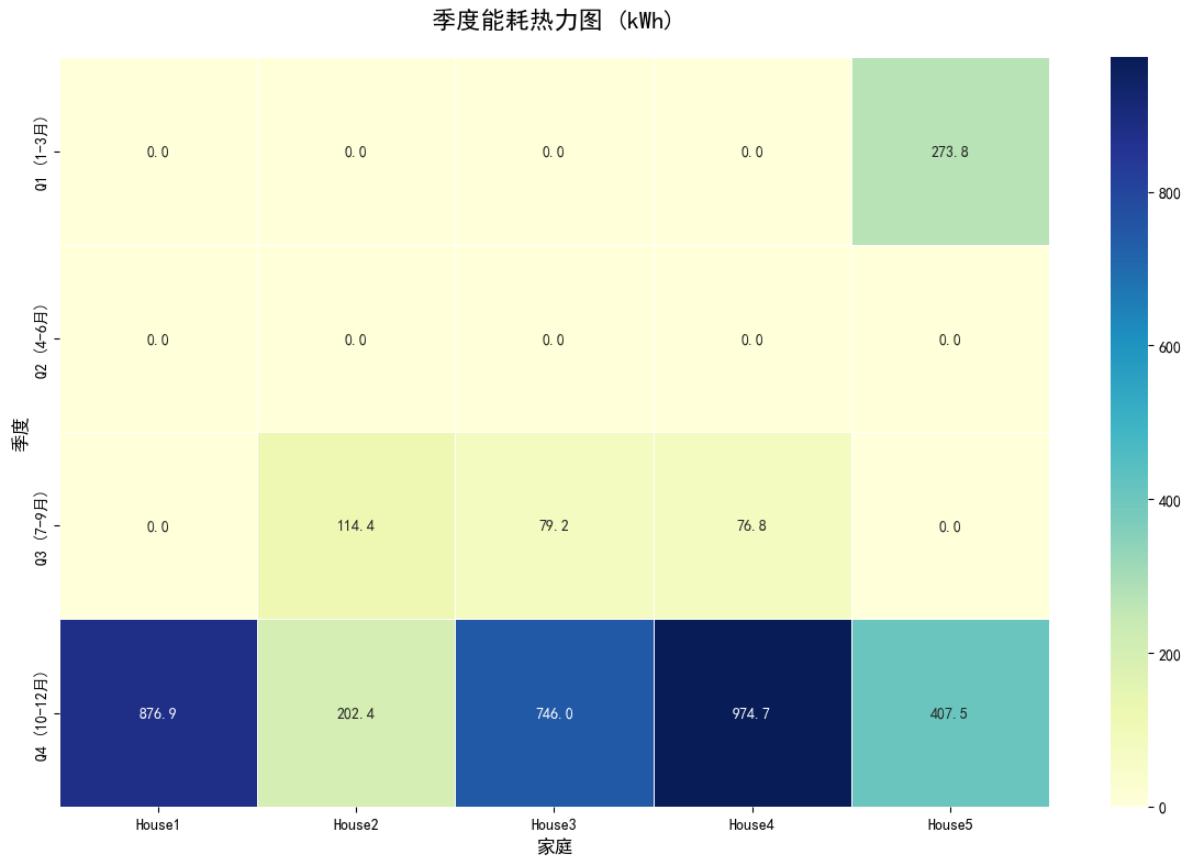
Appliance Stability: Dishwashers and washing machines showed consistent seasonal contributions, with House 2's dishwasher usage remaining at 14.06% of total consumption

year-round.

Heatmaps (Appendix Figure 3) visualized quarterly distribution, confirming Q4 as the dominant period for all households except House 5, which split consumption between Q1 (40.2%) and Q4 (59.8%)







4 Discussion and Insights

4.1 Time-of-Use (TOU) Pricing Models:

Electricity companies could leverage the hourly consumption peaks (7–9 AM, 18–21 PM) to design dynamic pricing structures. By setting higher rates during these peak windows (e.g., 20% higher tariffs from 18–21 PM) and lower rates during off-peak hours (10 AM–3 PM), they can incentivize households to shift non-urgent appliance use (e.g., dishwashing, laundry) to less strained periods. This directly addresses the dual daily peaks observed across all households, which strain grid capacity—for instance, House 4's 86.72 kWh peak at 18 PM, the highest hourly consumption in the dataset.

4.2 Grid Load Management:

Power utilities can use the quarterly trends (e.g., Q4 dominance in 4 out of 5 households) to proactively adjust grid resources. For example, increasing power generation or grid reinforcement in Q4 (October–December) to accommodate the 90.4–92.7% of annual consumption concentrated in this period (as seen in House 3 and House 4) would reduce the

risk of blackouts or voltage fluctuations. This leverages the seasonal patterns identified in the quarterly analysis, ensuring resource allocation matches actual demand.

4.3 Household Energy Management:

The hourly peaks (7–9 AM, 18–21 PM) and appliance-specific cycles (e.g., dishwashers in evenings, washing machines in mornings) can inform smart home systems. For instance, programmable timers could automatically delay non-urgent appliance use (e.g., dishwashing) to off-peak hours (10 AM–3 PM), reducing strain on household energy budgets without disrupting routines. This leverages the observed flexibility in appliance scheduling

4.4 Appliance Design and Marketing:

Manufacturers could refine product features based on usage patterns. For example, washing machines with enhanced "off-peak mode" (optimized for 10 AM–3 PM operation) or dishwashers with shorter cycles tailored to evening peak windows (18–21 PM) might better align with actual household behavior, as identified in the hourly activity stats (e.g., House 3's 5,919 morning washing machine activations)

5 Conclusion

This project successfully fulfilled the core objectives outlined in the assignment, applying data preprocessing and exploratory data analysis (EDA) techniques to real-world household power consumption data.

In the data cleaning phase, rigorous checks were conducted for common issues such as missing values, duplicates, and negative values. Despite the high quality of the dataset—with no missing entries, duplicates, or implausible negative values—the structured pipeline (including validation, correction, and verification steps) ensured data integrity, adhering to the requirement to justify cleaning methods . This process reinforced the importance of robust preprocessing in enabling reliable subsequent analysis.

The EDA phase uncovered meaningful patterns in household energy behavior. Temporal analysis revealed consistent daily peaks in aggregate consumption (7–9 AM and 18–21 PM), distinct activity cycles for dishwashers and washing machines, and seasonal trends with Q4 (October–December) as a dominant period for energy use. These findings, supported by visualizations such as line plots, heatmaps, and stacked charts, clarified how individual

appliances contribute to overall power consumption .

Looking forward, future work could extend this analysis by incorporating data from additional household appliances (e.g., refrigerators, air conditioners) to gain a more comprehensive view of energy dynamics. Integrating external variables such as weather data or household occupancy could also enhance the explanatory power of observed patterns.

6 References

6.1 Technical Tools & Methodologies

1. The Deepseek large model has been of great help when modifying programs, solving problems such as confusing dictionary usage references and errors in function calls one by one..
2. The Doubao large model plays a role in providing ideas for analyzing data.
3. Python data analysis presentation PPT and after-class assignments
4. Youdao Dictionary plays a significant role in the translation between Chinese and English when writing English papers